

Algebra II with Trigonometry

Chapter 7.2

Mental Math (Gemini)

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[4582337_alg-2trig-h-chp-7-2-note-docx.pdf](#)

Cosine Addition Formula 1. What is the exact numerical value of $\cos(40^\circ)\cos(20^\circ) - \sin(40^\circ)\sin(20^\circ)$?

Sine Subtraction Formula 2. Evaluate the following expression to a single number: $\sin(100^\circ)\cos(10^\circ) - \cos(100^\circ)\sin(10^\circ)$.

Tangent Addition Formula 3. Simplify the following fraction to a single whole number: $\frac{\tan(20^\circ) + \tan(25^\circ)}{1 - \tan(20^\circ)\tan(25^\circ)}$.

Cosine Subtraction in Radians 4. What is the exact value of $\cos\left(\frac{5\pi}{12}\right)\cos\left(\frac{\pi}{12}\right) + \sin\left(\frac{5\pi}{12}\right)\sin\left(\frac{\pi}{12}\right)$?

Amplitude of a Linear Combination 5. When rewriting the expression $3\sin(x) + 4\cos(x)$ into the single trigonometric form $k\sin(x + \phi)$, what is the value of k ?

Linear Combination with Radicals 6. If you express $-\sqrt{3}\sin(x) + \cos(x)$ in the form $k\sin(x + \phi)$, what is the resulting value of k ?

Co-function Identities 7. Using the addition/subtraction formulas in your head, what does the expression $\cos\left(x - \frac{\pi}{2}\right)$ simplify to?

Nature of Inverse Trigonometric Functions 8. According to the text's rules for evaluating inverse trigonometric functions, what basic geometric concept does an expression like $\cos^{-1}(x)$ represent?

Evaluating Inverse Trig Combinations 9. What is the exact value of $\sin(\sin^{-1}(1) + \cos^{-1}(1))$?

Tangent Subtraction Formula 10. What does the expression

$\frac{\tan(60^\circ) - \tan(15^\circ)}{1 + \tan(60^\circ) \tan(15^\circ)}$ evaluate to?